

AbRSA User Guide

Antibody sequence numbering and complementarity determining region (CDR) delimitation have wide applications in antibody engineering. They are generally obtained by mapping query sequences to the retrospective patterns. However, due to the enormous diversity of antibody sequences, novel patterns may not be recognized by the traditional methods. This method, namely Antibody Region-Specific Alignment (**AbRSA**), integrates the antibody region-specific feature with dynamic programming to improve the robustness of antibody numbering. Benchmarks showed that AbRSA was particularly robust in numbering unusual antibodies.

AbRSA supports three widely used numbering schemes, including **Chothia**, **Kabat**, and **IMGT** schemes. It can automatically determine whether the protein sequences are heavy chain, light chain or other sequences. Single chain antibodies such as nanobodies are also supported.

AbRSA was developed in **Yang Cao** Lab, Sichuan University. It is a command line program (x86-64) running on **Linux system**. AbRSA is freely available for academic users. Commercial users please contact us (cy_scu@yeah.net).

Please Cite: AbRSA: a Robust Tool for Antibody Numbering. ***Protein Science***. 28, 1524-1531, 2019

Usage: AbRSA -i [Query Antibody amino acid FASTA File]

Option: -c (using Chothia scheme, default)

-k (using Kabat scheme)

-g (using IMGT scheme)

-o [Output numbering file]

-z [similarity cutoff, default value = 70]

-h (help information)

Example:

(1)Numbering the antibody in Chothia scheme

```
$ ./AbRSA -i ab.fas -c -o ab_numbering.txt
```

```
#-----  
# AbRSA: a Tool for Antibodies Numbering and CDRs delimitation  
#           Yang Cao, All Rights Reserved.  
#           V 2.10  
#           Thu May 26 22:58:50 2022  
  
# Please Cite: AbRSA: a Robust Tool for Antibody Numbering.  
#           Protein Science. 28, 1524-1531, 2019  
#-----
```

```
# Delimiting CDRs in Chothia scheme.
```

```
>30AU-H
```

```
#similarity 88 %
```

```
H_FR1 : EVQLVESGGGLVKAGGSLRLSCGVS
```

```
H_CDR1: NFRISAH
```

```
H_FR2 : TMNWVRRVPGGGLWVASI
```

```
H_CDR2: STSSTY
```

```
H_FR3 : RDYADAVKGRFTVSRDDLEDFVYLQMHKMRVEDTAIYYCAR
```

```
H_CDR3: KGSDRLSDNDPFDA
```

```
H_FR4 : WPGGTVVTVSPA
```

```
-_EXT :
```

```
STKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVT
```

VPSSSLGTQTYICNVNHKPSNTKVDKKVEPKS

>1FBI-L

#similarity 93 %

L_FR1 : DIQMTQTTSSLSASLGDRVTISC

L_CDR1: RASQDISNYLN

L_FR2 : WYQKKPDGTVKLLIY

L_CDR2: YTSRLHS

L_FR3 : GVPSRFSGSGSGTDYSLTIRNLEQEDIATYFC

L_CDR3: QQGYTLPYT

L_FR4 : FGGGTKLEIK

-_EXT

RADAAPTVSIFPPSSEQLTSGGASVVCFLNNFYPKDINVKWKIDGSERQNGVLNSWTDQDSKDSTYSMS
STLTLTKEDEYERHNSYTCEATHKTSTSPIVKSFNRNEC

(2)Numbering the antibody in IMGT scheme

```
$ ./AbRSA -i ab.fas -g -o ab_numbering.txt
```

```
#-----  
# AbRSA: a Tool for Antibodies Numbering and CDRs delimitation  
#           Yang Cao, All Rights Reserved.  
#           V 2.10  
#           Thu May 26 23:08:23 2022  
  
# Please Cite: AbRSA: a Robust Tool for Antibody Numbering.  
#           Protein Science. 28, 1524-1531, 2019  
#-----
```

Delimiting CDRs in Imgt scheme.

>30AU-H

#similarity 87 %

H_FR1 : EVQLVESGGGLVKAGGSLRLSCGVS

H_CDR1: NFRISAHT

H_FR2 : MNWVRRVPGGGLEWVAS

H_CDR2: ISTSSTYR

H_FR3 : DYADAVKGRFTVSRDDLEDFVYLQMHKMRVEDTAIYYC

H_CDR3: ARKGSDRLSDNDPFDA

H_FR4 : WPGGTVVTVSP

-_EXT

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSV

TVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKS

>1FBI-L

#similarity 93 %

L_FR1 : DIQMTQTTSSLSASLGDRVTISCRAS

L_CDR1: QDISNY

L_FR2 : LNWYQKKPDGTVKLLIY

L_CDR2: YTS

L_FR3 : RLHSGVPSRFSGSGSGTDYSLTIRNLEQEDIATYFC

L_CDR3: QQGYTLPYT

L_FR4 : FGGGTKLEIK

-_EXT

:

RADAAPTVSIFPPSSEQLTSGGASVVCFLNNFYPKDINVKKWKIDGSERQNGVLNSWTDQDSKDSTYSMS

STLTLTKEDEYERHNSYTCEATHKTSTSPIVKSFNRNEC